

Attic and Crawl Space Ventilation

Two ratios, and your five-times miss about when soffit vents can be left out.

Ventilation works by letting cool air in low and warm air out high — the stack effect, with cool air entering at the eaves and warm air leaving at the ridge. Both ends are required for it to work at all.

The numbers here

Attic ventilation — standard requirement	1/300 of the attic floor area
Base attic ventilation requirement, and crawl space ventilation	1/150
Roof slope under 2:12, or a cathedral roof with no attic — recommended vent area	1/150 of the roof area
Crawl space vent openings — within this distance of each corner	3 ft

The base requirement is one square foot of net free ventilating area for every 150 square feet of floor. It halves to 1/300 when the ventilation is well balanced between high and low, or a vapor retarder is present. Where there is no attic to work with at all, it goes back to 1/150.

Net free area, not the size of the hole

A vent's net free area is the actual opening left after the louvres and the insect screen take their share — always less than the hole it sits in. When a ventilation question gives you an area, check whether it is asking for gross or net free.

YOU HAVE MISSED THIS 5 TIMES

When can soffit venting be omitted?

If gable vents are installed at opposing ends.

Five times missed. Soffit vents are the low inlet. You can leave them out only if something else provides cross-flow, and gable vents at opposing ends do that — air enters one gable and leaves the other, straight through.

YOU HAVE MISSED THIS 4 TIMES

Gable wall ventilation openings should be blocked when _____ ventilation is installed.

Ridge.

This looks like the opposite of the last one, and it is worth seeing why it is not. Ridge venting draws from the soffits, low to high. Leave the gable vents open as well and the ridge pulls its air from the nearest easy opening — the gables — instead of from the soffits, so the lower half of the attic gets no flow. Gables can substitute for soffits, but they cannot coexist with a ridge vent.

YOU HAVE MISSED THIS

Installing extra ventilation in an attic can have what effect?

It can cause negative pressure in the attic.

More exhaust than intake pulls the attic below the pressure of the house beneath it, which draws conditioned air up through every ceiling gap. More is not better; balanced is better.

YOU HAVE MISSED THIS

A crawl space always requires:

Access.

Say it back

Cover the page. Answer each one out loud.

1. Explain the stack effect in one sentence.
2. Give the attic ventilation ratios, and say what halves the requirement.
3. What ratio applies to a cathedral roof with no attic?
4. How close to each corner must crawl space vents sit?
5. What is net free area?
6. When may soffit venting be omitted?
7. When must gable openings be blocked, and why is that not a contradiction?
8. What can too much attic ventilation cause?